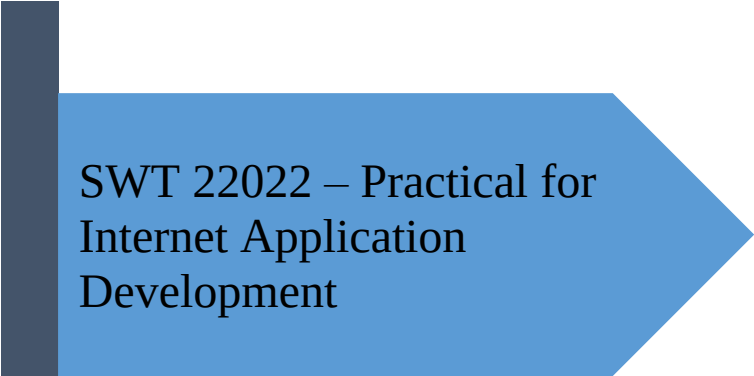




SWT 22022 – Practical for
Internet Application
Development

Department of Information
and Communication
Technology
Faculty of Technology



Lab sheet :07
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Practical No :07

Title: Getting and Passing Data from HTML to PHP Using Ajax.

Aims:

- To understand the concept of Ajax and its benefits.
- To learn how to make Ajax request with JQuery.
- To learn how to process Ajax request in PHP

Tasks:

- Import JQuery
- Implement AJAX request with JQuery
- Perform text field validation
- Process data in PHP

Activities :

1. Import JQuery

```
<scriptsrc="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
```

2. Implement AJAX request with JQuery

I. POST request

```
$.ajax({
    url: "add_student.php",
    type: "POST",
    data: {
        name: name,
        gpa: gpa
    },
    success: function(response) {
        try {
            var data = JSON.parse(response);
            if (data.success) {
                $("#studentForm").trigger("reset");
                console.log("Student added successfully!");
                if (data.students) {
                    $("#studentTable tbody").empty();
                    for (var i = 0; i < data.students.length; i++) {
                        var student = data.students[i];
                        var tableRow = "<tr>";
```

```

        tableRow += "<td>" +student.id + "</td>";
        tableRow += "<td>" +student.name +
    "</td>";
        tableRow += "<td>" +student.gpa +
        tableRow += "</tr>";
        $("#studentTable
tbody").append(tableRow);
    }
    } else {
        alert("Error adding student: " +data.message);
    }
} catch (error) {
    console.error("Error parsing response:", error);
    alert("An error occurred while adding student. Please check your
server logs.");
}
},
error: function (jqXHR, textStatus,errorThrown) {
    console.error("AJAX Error:", textStatus,errorThrown);
    alert("An error occurred while adding student.");
}
});

```

II. GET request

```

$.ajax({
    url: "get_students.php",
    type: "GET",
    success: function(response) {
        try {
            var data = JSON.parse(response);
            if (data.success && data.students &&
Array.isArray(data.students)) {
                $("#studentTable tbody").empty();
                for (var i = 0; i <
data.students.length;i++) {
                    var student = data.students[i];
                    var tableRow = "<tr>";
                    tableRow += "<td>" + student.id
+ "</td>";
                    tableRow += "<td>" + student.name
+ "</td>";
                    tableRow += "<td>" + student.gpa
+ "</td>";
                    tableRow += "</tr>";
                    $("#studentTable
tbody").append(tableRow);
                }
            }
        } catch (error) {
            console.error("Error parsing response:", error);
            alert("An error occurred while adding student. Please check your
server logs.");
        }
    },
    error: function (jqXHR, textStatus,errorThrown) {
        console.error("AJAX Error:", textStatus,errorThrown);
        alert("An error occurred while adding student.");
    }
});

```

```

        }
    } else {
        console.warn("Unexpected response
format:students array not found.");
    }
} catch (error) {
    console.error("Error parsing response:",error);
    alert("An error occurred while retrieving
student data.");
}
},
error: function(jqXHR, textStatus, errorThrown) {
    console.error("AJAX Error:", textStatus,errorThrown);
    alert("An error occurred while retrieving student
data.");
}
});

```

2. Perform text field validation

```

var name = $("#name").val();
var gpa = parseFloat($("#gpa").val());
if (name.trim() === "") {
    alert("Please enter a name.");
    return;
}
if (isNaN(gpa) || gpa < 0 || gpa > 4) {
    alert("Please enter a valid GPA between 0 and 4.");
    return;
}

```

4. Process data in PHP

I. get_students.php

```

if ($_SERVER['REQUEST_METHOD'] === 'GET') {
    include ('DatabaseConnection.php');
    include ('Student.php');
    $db = new DatabaseConnection("localhost", "root", "", "faculty");

```

```

        $connection = $db->getConnection();
        $student = new Student($connection);
        $studentData = $student->getAllStudent();
        $response = array();
        if ($studentData) {
            $response['success'] = true;
            $response['students'] = $studentData;
        } else {
            $response['success'] = false;
            $response['message'] = "Error creating
student";
        }
        echo json_encode($response);
    } else {
        echo "Invalid request.";
    }
}

```

II. add_students.php

```

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name = $_POST['name'];
    $gpa = floatval($_POST['gpa']);
    include ('DatabaseConnection.php');
    include ('Student.php');
    $db = new DatabaseConnection("localhost", "root", "", "faculty");
    $connection = $db->getConnection();
    $student = new Student($connection);
    $student->setName($name);
    $response = array();
}

```

```

        if ($result) {
            $response['success'] = true;
            $studentData = $student->getAllStudent();
            $response['students'] = $studentData;
        } else {
            $response['success'] = false;
            $response['message'] = "Error creating student";
        }
        echo json_encode($response);
    } else {
        echo "Invalid request.";
    }
}

```

Index.php

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Students</title>
    <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
QWTKZyjpPEjISv5WaRU90FeRpok6YctnYmDr5pNlyT2bRjXh0JMhJy6hW+ALEwIH" crossorigin="anonymous">
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
integrity="sha384-YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDz0xhy9GkcIdslK1eN7N6jIeHz"
crossorigin="anonymous"></script>
</head>
<body class="bg-light">

<div class="container mt-5">
    <h1 class="mb-4">Student Management</h1>

    <form id="studentForm" class="mb-4">
        <div class="mb-3">
            <label for="name" class="form-label">Student Name</label>
            <input type="text" id="name" class="form-control" placeholder="Student Name"
required>

```

```

    </div>
    <div class="mb-3">
        <label for="gpa" class="form-label">GPA</label>
        <input type="number" id="gpa" class="form-control" placeholder="GPA" required
step="0.01" min="0" max="4.0">
    </div>
    <input type="submit" value="Add Student" class="btn btn-primary">
</form>

<table id="studentTable" class="table table-striped">
    <thead>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>GPA</th>
        </tr>
    </thead>
    <tbody>
    </tbody>
</table>
</div>

<script>
$(document).ready(function() {
    loadStudents();

    // Prevent default form submission
    $("#studentForm").on("submit", function(event) {
        event.preventDefault(); // Prevent the form from submitting the traditional way
        addStudent(); // Call the function to add a student
    });
});

function loadStudents() {
    $.ajax({
        url: "get_students.php",
        type: "GET",
        success: function(response) {
            try {
                var data = JSON.parse(response);
                if (data.success && data.students && Array.isArray(data.students)) {
                    $("#studentTable tbody").empty();
                    for (var i = 0; i < data.students.length; i++) {
                        var student = data.students[i];
                        var tableRow = "<tr>";
                        tableRow += "<td>" + student.Id + "</td>";
                        tableRow += "<td>" + student.Name + "</td>";
                        tableRow += "<td>" + student.Gpa + "</td>";
                    }
                }
            }
        }
    });
}

```

```

        tableRow += "</tr>";
        $("#studentTable tbody").append(tableRow);
    }
    } else {
        console.warn("Unexpected response format: students array not found.");
    }
} catch (error) {
    console.error("Error parsing response:", error);
    alert("An error occurred while retrieving student data.");
}
},
error: function(jqXHR, textStatus, errorThrown) {
    console.error("AJAX Error:", textStatus, errorThrown);
    alert("An error occurred while retrieving student data.");
}
});
}

function addStudent() {
    var name = $("#name").val().trim();
    var gpa = parseFloat($("#gpa").val());

    if (name === "") {
        alert("Please enter a name.");
        return;
    }
    if (isNaN(gpa) || gpa < 0 || gpa > 4) {
        alert("Please enter a valid GPA between 0 and 4.");
        return;
    }

    $.ajax({
        url: "add_student.php",
        type: "POST",
        data: {
            name: name,
            gpa: gpa
        },
        success: function(response) {
            try {
                var data = JSON.parse(response);
                if (data.success) {
                    $("#studentForm").trigger("reset");
                    loadStudents(); // Reload the student list
                    console.log("Student added successfully!");
                } else {
                    alert("Error adding student: " + data.message);
                }
            }
        }
    });
}

```



```

        } catch (error) {
            console.error("Error parsing response:", error);
            alert("An error occurred while adding the student. Please check your server
logs.");
        }
    },
    error: function(jqXHR, textStatus, errorThrown) {
        console.error("AJAX Error:", textStatus, errorThrown);
        alert("An error occurred while adding the student.");
    }
});
});
}
</script>
</body>
</html>

```

Student.php

```

<?php
class Student {
    private $name;
    private $gpa;
    private $id;
    private $connection;

    public function __construct($connection) {
        $this->connection = $connection;
    }

    public function setName($name) {
        $this->name = mysqli_real_escape_string($this->connection, $name);
    }

    public function setGpa($gpa) {
        $this->gpa = $gpa;
    }

    public function create() {
        if (empty($this->name) || empty($this->gpa)) {
            return false;
        }
        $query = "INSERT INTO student (Name, Gpa) VALUES (?, ?)";
        $stmt = mysqli_prepare($this->connection, $query);
        mysqli_stmt_bind_param($stmt, 'sd', $this->name, $this->gpa);

        return mysqli_stmt_execute($stmt);
    }
}

```

```

    public function getAllStudents() {
        $query = "SELECT * FROM student";
        $result = mysqli_query($this->connection, $query);
        if (!$result) {
            return [];
        }
        $students = [];
        while ($row = mysqli_fetch_assoc($result)) {
            $students[] = $row;
        }
        return $students;
    }
}

?>

```

DatabaseConnection.php

```

<?php
class DatabaseConnection {
    private $host;
    private $username;
    private $password;
    private $database;
    private $connection;

    public function __construct($host, $username, $password, $database) {
        $this->host = $host;
        $this->username = $username;
        $this->password = $password;
        $this->database = $database;

        $this->connect();
    }

    private function connect() {
        $this->connection = mysqli_connect($this->host, $this->username, $this->password,
$this->database);
        if (!$this->connection) {
            die("Database connection failed: " . mysqli_connect_error());
        }
    }

    public function getConnection() {
        return $this->connection;
    }
}

?>

```

get_students.php

```
<?php
if ($_SERVER['REQUEST_METHOD'] === 'GET') {
    include ('DatabaseConnection.php');
    include ('Student.php');
    $db = new DatabaseConnection("localhost", "root", "", "faculty");
    $connection = $db->getConnection();
    $student = new Student($connection);
    $studentData = $student->getAllStudents();
    $response = array();
    if ($studentData) {
        $response['success'] = true;
        $response['students'] = $studentData;
    } else {
        $response['success'] = false;
        $response['message'] = "Error creating student";
    }
    echo json_encode($response);
} else {
    echo "Invalid request.";
}
```

add_student.php

```
<?php
if ($_SERVER['REQUEST_METHOD'] === 'POST') {

    $name = trim($_POST['name']);
    $gpa = floatval($_POST['gpa']);

    if (empty($name) || $gpa < 0 || $gpa > 4) {
        echo json_encode([
            'success' => false,
            'message' => 'Invalid input data. Please check the name and GPA.'
        ]);
        exit();
    }

    include ('DatabaseConnection.php');
    include ('Student.php');

    $db = new DatabaseConnection("localhost", "root", "", "faculty");
    $connection = $db->getConnection();
```

```

$student = new Student($connection);
$student->setName($name);
$student->setGpa($gpa);

$result = $student->create();

$response = array();
if ($result) {
    $response['success'] = true;
    $studentData = $student->getAllStudents();
    $response['students'] = $studentData;
} else {
    $response['success'] = false;
    $response['message'] = "Error creating student";
}

echo json_encode($response);
} else {
    echo json_encode(['success' => false, 'message' => 'Invalid request.']);
}
?>

```

Output:

Student Management

Student Name

GPA

[Add Student](#)

ID	Name	GPA
1	Pasan sawmya Kalhara	3.900
2	Pasan sawmya Kalhara	3.900
3	Avishaka Alok	2.900
4	Ishara Madushanka	4.000
5	Jasith Isiwara	1.900
6	Samitha Madhuranga	3.000
7	Githmal	3.000
8	Githmal	4.000

Discussion:

- **In this lab session, we aimed to gain a deeper understanding of how to transfer data between the client-side (HTML) and server-side (PHP) using AJAX. We began by learning the core benefits of AJAX, such as its ability to update web content asynchronously without requiring a full page reload. Using jQuery, we implemented AJAX requests, which simplified the process of sending and receiving data from the server. Additionally, we focused on validating text fields dynamically on the client side, ensuring that user inputs met specific criteria before sending the data. On the server-side, we processed the submitted data using PHP, demonstrating how AJAX facilitates seamless communication between the front-end and back-end, leading to faster interactions and a smoother user experience. This integration of jQuery and PHP through AJAX highlighted how modern web applications can achieve real-time validation and data processing.**